WEN, HSING-YU (Winnie)

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EDUCATION

Carnegie Mellon University, Pennsylvania — August 2025 - December 2026
Master of Information Systems Management - Business Intelligence and Data Analytics
Soochow University, Taiwan — September 2019 – June 2023
Bachelor of Business Administration (BBA) in International Business

- **GPA**: 4.00/4.00 (Class Rank: 1/68; Department Rank: 1/136)

SKILLS & LANGUAGES

- Programming & Tools: Python, SQL, R, Tableau, Java, Power BI, Excel, SPSS, Git, GitHub, Docker, Kafka
- Data Analytics & Machine Learning: Regression Analysis, A/B Testing, Predictive Modeling, Machine Learning, scikit-learn, TensorFlow, PyTorch, NLP, Large Language Models (LLMs), Model Deployment, Model Evaluation, MLOps, Monitoring
- Database & Cloud: MSSQL, MongoDB, AWS, Hadoop, Airflow, Databricks

WORK EXPERIENCE

TSMC, Hsingchu City, Taiwan — May 2026 - August 2026
Data AI Engineer

- Built an enterprise AI knowledge platform indexing 5,000+ Power BI reports by combining MariaDB for structured metadata and Qdrant for semantic indexing, enabling Retrieval-Augmented Generation for natural language analytics.
- Built an LLM-powered knowledge generation pipeline that transforms Power BI metadata extracted via REST and Scanner APIs into standardized AI-ready knowledge assets (e.g., report purpose, key visualizations, and business terminology), saving over 1,200 hours of manual documentation effort.
- Developed retrieval pipelines that map natural language queries to the most relevant Power BI datasets, integrating with internal Model Context Protocol (MCP) services to support downstream AI-generated SQL/DAX workflows.

Bank Sinopac, Taipei City, Taiwan — June 2024 - June 2025
Data Analyst

- **Data Visualization:** Engineered 20+ enterprise-grade interactive dashboards leveraging Python, T-SQL, and Tableau, streamlining decision-making workflows by 47% and elevating data accessibility across business units
- **Machine Learning:** Architected uplift models and validated strategies via A/B testing, driving marketing ROI by 25% and delivering actionable insights for senior leadership
- **Business Impact:** Lead Data Analyst for 2025 flagship OMO project, enabling relationship managers to engage high-potential online customers and driving NT$5–10M monthly transactions

Asus, Taipei City, Taiwan — May 2023 - November 2023
Data Analyst Intern

- **Statistical Modeling:** Executed logistic regression to identify industry-specific content preferences, refining campaign targeting and outperforming previous campaign conversion rates by 31%

ACADEMIC PROJECTS

Banking Text-to-SQL RAG Project — May 2026 - Ongoing

- **LLM Systems / RAG / Text-to-SQL**: Designed and implemented a production-inspired Text-to-SQL RAG workflow for banking data, incorporating automated schema indexing, semantic retrieval, context filtering, SQL safety validation, out-of-scope guardrails, and natural-language response generation to support reliable business-user access to structured data.

Movie Recommendation System | Streaming Data Pipeline & Recommendation Service — January 2026 - April 2026

- **Data Engineering:** Built an end-to-end streaming data pipeline integrating user/movie APIs with Kafka event logs, supporting 2,000+ recommendation requests within a 24-hour window for feature generation and model input
- **AI Engineering (Online Inference & Serving):** Designed a hybrid architecture (offline training + online inference API), deploying a containerized recommendation service to deliver personalized results under a 600ms latency constraint
- **MLOps:** Implemented data quality checks, pipeline testing, and monitoring, including drift detection and offline/online evaluation metrics (e.g., HitRate@K), ensuring reliable performance in a production-like environment

Text Mining of Chimei Museum Google Reviews — October 2022 - January 2023

- **Unstructured Data Analysis:** Developed a scalable web scraping and NLP pipeline processing 50K+ Google reviews; applied TF-IDF and LDA topic modeling to extract 5 sentiment themes, boosting stakeholder understanding of customer feedback by 40% and guiding engagement strategies